Unit 2. Introduction to ASP.NET 3 Hrs.

.NET and ASP.NET frameworks: .NET, .NET Core, Mono, ASP.NET Web Forms, ASP.NET MVC, ASP.NET Web API, ASP.NET Core, .NET Architecture and Design Principles, Compilation and Execution of .NET applications: CLI, MSIL and CLR, .NET Core in detail, .NET CLI: build, run, test and deploy .NET Core Applications

Unit 3. HTTP and ASP.NET Core 3 Hrs.

HTTP, Request and Response Message Format, Common web application architectures, MVC Pattern, ASP.NET Core Architecture Overview, Projects, and Conventions, ASP.NET and ASP.NET MVC

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.NET and ASP.NET Frameworks Overview

- **.NET Framework:** Windows-only development platform.
- **.NET Core:** Cross-platform, open-source version of .NET.
- **Mono**: Open-source implementation of .NET for Linux and mobile platforms.
- **ASP.NET Web Forms:** Event-driven web development using server controls.
- **ASP.NET MVC:** Web framework based on the Model-View-Controller architecture.
- **ASP.NET Web API**: Framework for building RESTful web services.
- **ASP.NET Core:** Modern cross-platform framework for web applications and APIs.

1. .NET Framework

.NET Framework is Microsoft's software development platform used to build Windows desktop applications, web applications, and services. It provides a large library (Framework Class Library - FCL) and a runtime called CLR (Common Language Runtime).

Features

- *Runs mainly on Windows.*
- *Supports languages like C#, VB.NET, and F#.*
- *Provides automatic memory management (Garbage Collection).*

```
using System;

class Program
{
    static void Main()
    {
        Console.WriteLine("Hello .NET Framework!");
    }
}
```

Output

```
Hello .NET Framework!
```

2. .NET Core

.NET Core is a modern, lightweight, open-source, and cross-platform version of .NET. It runs on Windows, Linux, and macOS.

Features

- *Cross-platform*
- *Faster performance*
- *Open source*
- *Ideal for cloud applications and microservices*

```
using System;

Console.WriteLine("Hello .NET Core!");
```

Output

```
Hello .NET Core!
```

Example Use

- REST APIs
- Cloud applications
- Cross-platform software

3. Mono

Mono is an open-source implementation of Microsoft's .NET Framework that allows .NET applications to run on Linux, macOS, Android, and iOS.

Features

- *Cross-platform*
- *Supports mobile development*
- *Used before .NET Core became popular*

Example Use

- Running a C# application on Linux.

```
using System;

class Program
{
    static void Main()
    {
        Console.WriteLine("Running with Mono");
    }
}
```

Output

```
Running with Mono
```

4. ASP.NET Web Forms

ASP.NET Web Forms is Microsoft's event-driven framework for creating web applications. It works similarly to Windows Forms using drag-and-drop controls.

Features

- *Event-driven programming*
- *Server controls (Button, TextBox, GridView)*
- *Easy for beginners*

Output

When the button is clicked:

```
Welcome to ASP.NET Web Forms
```

Default.aspx

‹/› aspx

```
<asp:Button ID="btnHello"
    runat="server"
    Text="Click Me"
    OnClick="btnHello_Click" />

<asp:Label ID="lblMessage" runat="server"></asp:Label>
```

Default.aspx.cs

‹/› C#

```
protected void btnHello_Click(object sender, EventArgs e)
{
    lblMessage.Text = "Welcome to ASP.NET Web Forms";
}
```

5. ASP.NET MVC

ASP.NET MVC (Model-View-Controller) *is a framework that separates an application into three parts:*

- ***Model*** *→ Data*
- ***View*** *→ User Interface*
- ***Controller*** *→ Business Logic*

This separation makes applications easier to maintain and test.

Controller

```
</> C#

public class HomeController : Controller
{
    public IActionResult Index()
    {
        return Content("Welcome to ASP.NET MVC");
    }
}
```

Output

```
Welcome to ASP.NET MVC
```

6. ASP.NET Web API

ASP.NET Web API is a framework for building RESTful APIs that allow applications to exchange data using HTTP.

Features

- *Returns JSON or XML*
- *Used by web, mobile, and desktop applications*
- *Lightweight communication*

```
[ApiController]
[Route("api/hello")]
public class HelloController : ControllerBase
{
    [HttpGet]
    public string Get()
    {
        return "Hello from Web API";
    }
}
```

If we visit:

```
https://localhost/api/hello
```

Output

```
Hello from Web API
```

7. ASP.NET Core

ASP.NET Core is the latest, open-source, cross-platform framework for building web applications, REST APIs, and cloud-based applications. It combines the best features of MVC and Web API.

Features

- *Cross-platform*
- *High performance*
- *Open source*
- *Supports MVC, Razor Pages, Web API, and Blazor*
- *Runs on Windows, Linux, and macOS*

```
var builder = WebApplication.CreateBuilder(args);
var app = builder.Build();

app.MapGet("/", () => "Hello ASP.NET Core!");

app.Run();
```

Open browser:

```
https://localhost:5001/
```

Output

```
Hello ASP.NET Core!
```

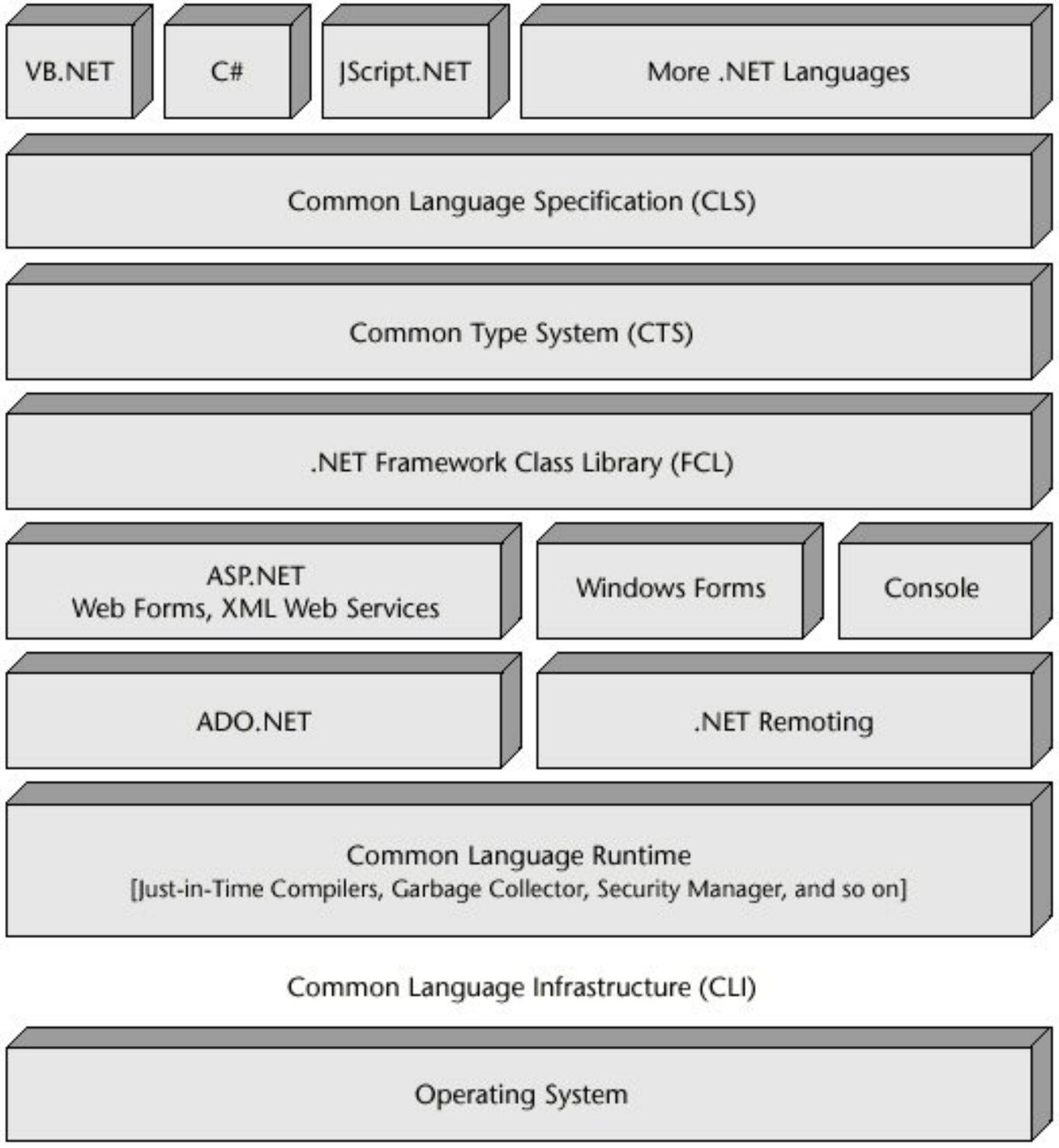
Technology	Platform	Main Purpose	Example Use
.NET Framework	Windows only	Desktop & Web applications	Windows Forms, WPF
.NET Core	Windows, Linux, macOS	Cross-platform applications	REST APIs, Cloud apps
Mono	Cross-platform	Run .NET apps on Linux/Mobile	Xamarin, Linux apps
ASP.NET Web Forms	Windows	Event-driven web applications	Company websites
ASP.NET MVC	Windows	Web applications using MVC pattern	E-commerce websites
ASP.NET Web API	Windows	RESTful APIs	Mobile app backend
ASP.NET Core	Cross-platform	Modern web apps & APIs	Cloud applications, Microservices

.NET Architecture and Design Principles

The .NET Architecture defines how a .NET application is organized and executed. It consists of languages (such as C#), the Common Language Runtime (CLR), the Framework Class Library (FCL), and the operating system.

Components

- ***Programming Language:*** *C#, VB.NET, F#*
- ***Compiler:*** *Converts source code into MSIL.*
- ***CLR:*** *Executes the program.*
- ***FCL (Framework Class Library):*** *Provides built-in classes (String, List, File, etc.).*
- ***Operating System:*** *Windows, Linux, or macOS.*



1. Programming Languages (Top Layer)

- **VB.NET, C#, JScript.NET, More .NET Languages:** These are the languages developers use to write code. The key feature here is that all these languages compile down to the same intermediate code, allowing them to work together seamlessly.

2. Language & Type Standards

- **Common Language Specification (CLS):** This is a set of rules that ensures language interoperability. It guarantees that code written in one language (e.g., C#) can use libraries written in another (e.g., VB.NET).
- **Common Type System (CTS):** This defines how data types are declared, used, and managed in the runtime. It ensures that an "integer" in C# is treated exactly the same as an "integer" in VB.NET.

3. The Library Layer

- **.NET Framework Class Library (FCL):** A massive collection of reusable classes, interfaces, and value types. It provides pre-written code for common tasks like file handling, string manipulation, and networking, so developers don't have to reinvent the wheel.

4. Application Models & Technologies

This layer represents the different types of applications and technologies you can build using the framework:

- **ASP.NET (Web Forms, XML Web Services):** Used for building dynamic websites, web applications, and web services.
- **Windows Forms:** Used for building traditional desktop applications with a Graphical User Interface (GUI).
- **Console:** Used for building command-line applications.
- **ADO.NET:** A set of components for data access. It allows applications to connect to data sources like SQL Server, Oracle, or XML files.
- **.NET Remoting:** A technology for communication between different application domains (allowing objects to interact across networks). *Note: This is an older technology largely replaced by WCF and Web APIs in modern .NET.*

5. The Execution Engine

- **Common Language Runtime (CLR):** This is the heart of the .NET Framework. It is the engine that actually runs your code.
    - **Just-in-Time (JIT) Compilers:** Converts the intermediate code into native machine code right before it runs.
    - **Garbage Collector:** Automatically manages memory by cleaning up objects that are no longer in use.
    - **Security Manager:** Enforces security policies.

6. Infrastructure & OS (Bottom Layer)

- **Common Language Infrastructure (CLI):** This is the international standard (ECMA/ISO) that describes the executable code and runtime environment. The CLR is Microsoft's implementation of the CLI.
- **Operating System:** The base layer (Windows, Linux, macOS) on which the entire framework runs. The CLI allows .NET to be portable across different operating systems.

Design Principles

1. Language Independence

Programs can be written in different .NET languages.

Example

- Module A → C#
- Module B → VB.NET
- Both can work together.

2. Platform Independence (.NET Core/.NET)

Run the same application on:

- Windows
- Linux
- macOS

3. Object-Oriented Programming

Supports:

- Encapsulation
- Inheritance
- Polymorphism
- Abstraction

4. Automatic Memory Management

CLR automatically removes unused objects using **Garbage Collection (GC)**.

C#

```
Student s = new Student();
```

When `s` is no longer used, CLR automatically frees its memory.

5. Security

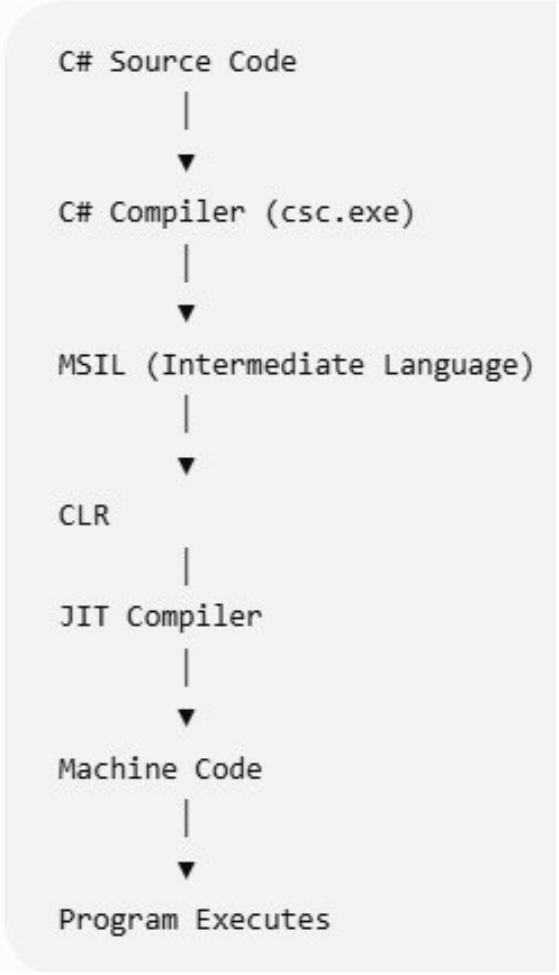
CLR provides:

- Type safety
- Exception handling
- Memory safety

.NET Architecture *is the overall structure of the .NET platform consisting of programming languages, the CLR, Framework Class Library (FCL), and the operating system. It provides language independence, security, portability, and automatic memory management.*

Compilation and Execution of .NET Applications

When we write a C# program, it is not executed directly. It goes through several stages.



(a) CLI (Common Language Infrastructure)

***CLI** is an international standard (ECMA/ISO) that defines how .NET languages and runtimes should work together. It specifies the rules for the runtime, intermediate language, metadata, and language interoperability.*

Main Components of CLI

- ***CTS (Common Type System)** – Defines common data types (int, string, bool, etc.).*
- ***CLS (Common Language Specification)** – Rules that all .NET languages follow to ensure interoperability.*
- ***MSIL (Microsoft Intermediate Language)** – Intermediate code generated by the compiler.*
- ***CLR (Common Language Runtime)** – Executes the program.*

Example
A class written in VB.NET can be used by a C# application because both follow the CLI standard.

(b) MSIL (Microsoft Intermediate Language)

MSIL (also called IL/CIL) is the intermediate code generated by the C# compiler. It is CPU-independent and cannot run directly.

MSIL is converted to machine code only when the program runs.

Example

Source Code

```
C#

Console.WriteLine("Hello");
```

After compilation

```
Hello.exe
   |
Contains MSIL code
```

(c) CLR (Common Language Runtime)

CLR is the execution engine of .NET. It converts MSIL into machine code using the Just-In-Time (JIT) compiler and provides services such as memory management, security, exception handling, and garbage collection.

Responsibilities of CLR

- Executes programs
- JIT compilation
- Memory management
- Garbage collection
- Exception handling
- Security
- Thread management

```
MSIL
 ↓
CLR
 ↓
JIT Compiler
 ↓
Machine Code
 ↓
Output:
Welcome
```

Example

```
C#

Console.WriteLine("Welcome");
```

JIT (Just-In-Time Compiler)

The JIT compiler converts MSIL into machine code only when the code is executed.

```
Console.WriteLine("Hello");
```

The first time this line runs:

```
MSIL
  ↓
JIT Compiler
  ↓
Machine Code
```

The generated machine code is then executed.

.NET Framework vs .NET Core

Feature	.NET Framework	.NET Core (.NET)
Platform	Windows only	Windows, Linux, macOS
Open Source	No	Yes
Performance	Good	Faster
Best For	Windows desktop applications	Web, cloud, APIs, microservices
Deployment	System-wide installation	Side-by-side versions supported

.NET CLI (Command Line Interface)

The .NET CLI (Command Line Interface) is a command-line tool used to create, build, run, test, publish, and deploy .NET applications without using Visual Studio. It is available on Windows, Linux, and macOS.

.NET CLI lets developers manage .NET projects using terminal commands.

1. `dotnet build` – Build the Application

The **build** command compiles the source code and checks for errors. It creates executable files (`.dll` or `.exe`) in the **bin** folder.

Syntax

Bash

```
dotnet build
```

Example

Bash

```
dotnet build MyApp.csproj
```

Output

```
Build succeeded.
0 Warning(s)
0 Error(s)
```

Use: Before running or deploying an application.

2. `dotnet run` – Run the Application

The **run** command builds (if necessary) and immediately executes the application.

Syntax

```bash
dotnet run
```

Example

Program.cs

```csharp
using System;

Console.WriteLine("Welcome to .NET Core!");
```

Command

```bash
dotnet run
```

Output

```
Welcome to .NET Core!
```

Use: Quickly test the application during development.

3. `dotnet test` – Test the Application

The **test** command runs unit tests written using frameworks like **xUnit**, **NUnit**, or **MSTest**.

Syntax

Bash

```
dotnet test
```

```
using Xunit;

public class CalculatorTest
{
    [Fact]
    public void AddTest()
    {
        Assert.Equal(5, 2 + 3);
    }
}
```

Command

Bash

```
dotnet test
```

Output

```
Passed! 1 test passed.
```

Use: *Verify that the application works correctly before deployment.*

4. `dotnet publish` – Deploy the Application

To deploy a .NET application, we use the **publish** command. It creates all files needed to run the application on another computer.

Syntax

```bash
dotnet publish
```

Example

```bash
dotnet publish -c Release
```

Output

```
Publishing completed.
Files generated in:
bin/Release/net8.0/publish/
```

The **publish** folder contains:

- Executable file (.exe)
- DLL files
- Configuration files
- Runtime dependencies

These files can be copied to a server or another computer for deployment.

Complete Example Workflow

Suppose you have a project named **HelloWorld**.

Step 1: Create a project

Bash

```
dotnet new console -n HelloWorld
```

Step 2: Move into the project

Bash

```
cd HelloWorld
```

Step 3: Build

Bash

```
dotnet build
```

Step 4: Run

Bash

```
dotnet run
```

Output:

```
Hello, World!
```

Step 5: Test (if test project exists)

Bash

```
dotnet test
```

Step 6: Publish for deployment

Bash

```
dotnet publish -c Release
```

Published files will be available in:

```
bin/
 └── Release/
      └── net8.0/
           └── publish/
```